

Heavy-Tailedness of Returns: Calibrating the ESN–A2 Architecture’s Excess Kurtosis via m_1

Companion note to the ESN A2-981003 market-data hyperparameter search

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Abstract

The companion note’s own diagnostic (§3.4) reports excess kurtosis – Fisher’s g_2 , the fourth standardised moment of daily log-returns minus 3, the standard measure of how heavy-tailed a return distribution is relative to a Gaussian – as systematically undershot by the optimal ESN–A2 architecture. This note evaluates and calibrates against excess kurtosis directly on the same 9-asset bench used throughout this note family (6 stocks, 3 indices).

The undershoot is severe and uniform: the companion note’s own classical (fixed centre p) architecture gives a kurtosis distribution of mean 2.79 (std 1.10, 60 replicate paths), while every one of the 9 real assets sits between $z = +2.9$ and $+12.6$ above it – among the largest, most consistent mismatches found across this whole note family.

A direct sweep at the same parameter region that resolved the leverage-effect, Taylor-effect, and Zumbach-effect notes’ own remaining gaps ($H = 0.3$, `rough_scale` = 0.7, $\lambda_{lo} \approx 0$, $\lambda_{hi} = 1.0$) already raises kurtosis from 2.79 to 29.9 on its own – comfortably above every real target. Unlike those 3 other notes, however, widening H or λ_{hi} further moves kurtosis in the *wrong* direction here; m_1 (the quadratic reservoir-norm coupling in η_t , held at 0 throughout every other calibration in this note family) is instead the parameter with a clean, monotonic effect: more positive raises kurtosis further, more negative lowers it. **Fixing the other 4 parameters at this shared corner and calibrating m_1 alone, per asset, closes the gap for every one of the 9 assets:** a 90% confidence band (60 replicate paths) covers the real value for all 9, with no exceptions. **A quantile-quantile check of the full return distribution, not just its 4th moment, shows this match holds well through the centre and most quantiles, but reveals an asymmetric tail mismatch for several assets** (most visibly SP500 and HUM) – a fatter-than-real left tail even where the overall kurtosis, and the right tail specifically, are well matched.

Contents

1	Motivation and set-up	2
2	The classical architecture’s own undershoot	2
3	Explicitly explaining the calibration	3
3.1	A first attempt, and why it did not work	3
3.2	How the non-calibrated parameters are fixed	3
3.3	The fix: fix 4 parameters, solve for m_1 alone	4
3.3.1	Why m_1 , not m_2 : a direct comparison	4
3.4	Confirmation and confidence band	5

4 Results	5
4.1 Excess kurtosis: real vs. calibrated ESN	5
4.2 Beyond kurtosis: quantile-quantile comparison of the full return distribution	7
4.3 The full return distribution, directly	9
5 Interpretation	10
6 Limitations and Recommended Next Steps	10

1 Motivation and set-up

Excess kurtosis (Eq. 1 below) is one of the companion note’s 11 stylised facts, the standard measure of how heavy-tailed a return distribution is relative to a Gaussian. This note evaluates and calibrates using the **same 9-asset bench as the leverage-effect, Taylor-effect, and Zumbach-effect notes**: 6 individual stocks (LHX, BK, HUM, TSN, IEX, KMB, seed 2026) and 3 major U.S. indices (SP500, RUSSELL2000, NASDAQ100).

$N_r = 22, N_z = 29$ and the 7 shared z -bank hyperparameters are fixed at the companion note’s own optimum throughout. Excess kurtosis itself is defined, for a return series x_t of length T with sample mean $\bar{x}$, as

$$g_2 = \frac{\frac{1}{T} \sum_{t=1}^T (x_t - \bar{x})^4}{\left(\frac{1}{T} \sum_{t=1}^T (x_t - \bar{x})^2\right)^2} - 3, \quad (1)$$

the bias-corrected 4th standardised moment (Fisher’s g_2) minus 3, so that a Gaussian series has $g_2 = 0$ and heavier-than-Gaussian tails give $g_2 > 0$.

2 The classical architecture’s own undershoot

Table 1: Real excess kurtosis, all 9 assets, vs. the classical (fixed centre p) architecture’s own distribution (mean 2.794, std 1.104, 60 replicate $T = 3,000$ -day paths).

Asset	type	real excess kurtosis	z -score
LHX	stock	8.578	+5.24
BK	stock	16.738	+12.63
HUM	stock	14.072	+10.21
TSN	stock	11.935	+8.28
IEX	stock	6.204	+3.09
KMB	stock	9.138	+5.74
SP500	index	10.314	+6.81
RUSSELL2000	index	7.724	+4.46
NASDAQ100	index	6.040	+2.94

Every one of the 9 assets sits well above the classical architecture’s own distribution, with no exceptions and no clean stock/index split – z -scores range from +2.9 to +12.6, among the most severe and uniform mismatches found across this whole note family.

3 Explicitly explaining the calibration

3.1 A first attempt, and why it did not work

An initial attempt jointly calibrated all 5 candidate parameters,

$$\theta = (H, \text{rough_scale}, \lambda_{lo}, \lambda_{hi}, m_1), \quad \theta \in \Theta \equiv [0.01, 0.3] \times [0.2, 0.7] \times [10^{-5}, 0.1] \times [1.0, 5.0] \times [-0.1, 0.1], \quad (2)$$

via bounded Levenberg–Marquardt. For a candidate θ , the ESN’s own estimate of g_2 is a Monte-Carlo average over $K = 6$ independent simulated paths ($T = 3,000$ days each, distinct seeds $s_1, \dots, s_K$), each computed by Eq. 1 on that single path:

$$\hat{g}_{2\theta} = \frac{1}{K} \sum_{k=1}^K \hat{g}_{2\theta}^{(s_k)}, \quad (3)$$

and the calibration attempted to solve the single-residual least-squares problem

$$\theta^* = \arg \min_{\theta \in \Theta} r(\theta)^2, \quad r(\theta) = \hat{g}_{2\theta} - g_2^{\text{real}}, \quad (4)$$

via `scipy.optimize.least_squares` (`method="trf"`, `diff_step=0.15`, up to 30 evaluations). **This got stuck:** starting from the same corner that resolved the other 3 notes’ own gaps ($H = 0.3$, `rough_scale`= 0.7, $\lambda_{lo} = 10^{-5}$, $\lambda_{hi} = 1.0$, all already at their own bounds), every asset converged to essentially the same point regardless of its own target, with m_1 barely moving from its starting value. **This is a genuine optimisation difficulty, not evidence the target is unreachable:** excess kurtosis is a high-variance statistic, dominated by rare, extreme observations, so the finite-difference gradient bounded LM estimates for m_1 is noisy relative to its true underlying sensitivity, and with the other 4 parameters already pinned at their own bounds from the start, 5-dimensional gradient search has no other direction to explore.

3.2 How the non-calibrated parameters are fixed

It is worth being precise about where every fixed value in this note actually comes from, since this note fixes more of p than any other in this family: only m_1 is calibrated per asset here, with $H, \text{rough_scale}, \lambda_{lo}, \lambda_{hi}$ – calibrated per asset in every other note in this family – also fixed, at a single value shared across every asset (§3.3).

$H, \text{rough_scale}, \lambda_{lo}, \lambda_{hi}$ are fixed at $(0.3, 0.7, 10^{-5}, 1.0)$ specifically because this is the same corner of the companion note’s own wide grid range for these 4 entries that resolved the leverage-effect, Taylor-effect, and Zumbach-effect notes’ own remaining gaps – not an arbitrary or unexamined choice, but one directly tested here too (§2, §3.3): it alone raises kurtosis from 2.79 to 29.9, and §3.1’s own attempt to also calibrate these 4 per asset showed the resulting 5-dimensional search could not move usefully away from this corner regardless of target.

The remaining 7 entries of $p - a_{lo}, a_{hi}, \text{zr_lo}, \text{zr_hi}, m_2, \Delta b_0, \text{scale}$ – are held fixed at the same values used throughout this whole note family, and it is worth being precise about where *those* come from too, since it is easy to assume they were individually optimised, and they were not. The companion note’s own architecture search draws a sharp distinction between *hyperparameters* (the architecture: N_r, N_z , the 7 shared z -bank values), which *were* found by bounded Levenberg–Marquardt against real market data, and *parameters* (p , all 12 entries, including every entry calibrated anywhere in this note family), which were not individually optimised at all. Instead, a 24-point randomised 2-level factorial design sampled every one of the 12 entries of p at either its

own low or high bound, and the resulting stylised facts were averaged across that whole grid – used only to give the architecture search a stable target, never to select a single best value for any one entry.

The 7 remaining fixed values are simply the midpoint of each entry’s own grid range from that search, not a value chosen or justified for excess kurtosis specifically:

Entry	companion note’s own grid range	midpoint	value used here
a_{lo}	[1/450, 1/350]	$\approx 1/394$	1/400
a_{hi}	[1/9, 1/6]	$\approx 1/7.2$	1/7
<code>zr_lo</code>	[0.01, 0.1]	0.055	0.055
<code>zr_hi</code>	[0.1, 0.4]	0.25	0.25
m_2	[−0.1, 0.1]	0	0
Δb_0	[−0.02, 0.02]	0	0
scale	[0.98, 1.02]	1.0	1.0

These midpoints were never individually revisited for excess kurtosis, or any other specific stylised fact – they are inherited as a convenient default from a search that used them only for marginalisation. m_2 is the one entry with a direct, if brief, test here: a sweep at the fixed corner found its own effect on kurtosis much weaker than m_1 ’s (a brief sweep, noted in Limitations below), so its midpoint value of 0 is at least a checked default, unlike the other 6, which were never examined for this stylised fact at all.

3.3 The fix: fix 4 parameters, solve for m_1 alone

3.3.1 Why m_1 , not m_2 : a direct comparison

η_t , the reservoir’s pre-activation (the companion note’s own definition; the leverage-effect note’s own Eq. 1 gives it in full), contains 2 quadratic terms, $m_1 \|r_t\|^2 / N_r$ and $m_2 (q^\top r_t)^2 / N_r$ – m_1 couples to the reservoir’s own total norm, m_2 specifically to its projection along q , the same direction the roughness channel itself uses. Both are candidate levers for kurtosis; a matched sweep at the fixed corner ($H = 0.3$, `rough_scale` = 0.7, $\lambda_{lo} = 10^{-5}$, $\lambda_{hi} = 1.0$), each point averaged over the same $K = 20$ paths ($T = 4,000$ days) used throughout this section, shows why m_1 was chosen:

m_1 ($m_2 = 0$)	−0.10	−0.05	0.00	+0.05	+0.10
$\widehat{g}_2$	7.32	9.64	13.19	18.55	26.40
m_2 ($m_1 = 0$)	−0.10	−0.05	0.00	+0.05	+0.10
$\widehat{g}_2$	12.18	12.67	13.19	13.73	14.30

m_1 moves kurtosis roughly 9× more than m_2 over the same range (a swing of 19.08 across $m_1 \in [-0.1, 0.1]$, against 2.12 for m_2 across the identical range) – $\partial \widehat{g}_2 / \partial m_1 \approx 95$ near $m_1 = 0$, vs $\partial \widehat{g}_2 / \partial m_2 \approx 11$. This is consistent with m_1 coupling to the reservoir’s *entire* state, while m_2 only reaches the single direction q already governs elsewhere in η_t – m_1 has substantially more of the reservoir’s own dynamics available to amplify, which is exactly what a heavy-tail-producing lever needs. m_2 is correspondingly left fixed at its own midpoint value throughout this note (§3.2), and m_1 is the parameter this section develops into the calibration itself:

m_1	−0.09	−0.05	0.00	+0.05	+0.09
$\widehat{g}_2$	15.33	20.48	29.91	44.16	60.42

(a single-path-per-point sweep, shown here at the original, coarser resolution used to first identify the effect, and consistent in direction and rough magnitude with the more heavily-averaged comparison above – the 2 sweeps differ somewhat in their exact values because of kurtosis’s own high estimation variance, discussed throughout this note, not because either is wrong) – exactly the kind of well-behaved 1-dimensional problem bisection solves reliably where 5-dimensional gradient descent did not. This note therefore fixes $H, \text{rough_scale}, \lambda_{lo}, \lambda_{hi}$ at this shared corner for every asset and solves

$$m_1^* : \quad \widehat{g}_2(m_1^*) = g_2^{\text{real}}, \quad m_1 \in [-0.1, 0.1], \quad (5)$$

per asset via bounded bisection (`scipy.optimize.brentq`), where, as in Eq. 3, $\widehat{g}_2(m_1)$ is itself a Monte-Carlo average, now over $K = 20$ independent simulated paths at a fixed, non-incrementing set of seeds $s_1, \dots, s_K$ ($T = 4,000$ days each):

$$\widehat{g}_2(m_1) = \frac{1}{20} \sum_{k=1}^{20} \widehat{g}_2^{(s_k)}(m_1), \quad (6)$$

substantially more replicates than Eq. 3’s own $K = 6$, and critically *fixed* rather than freshly drawn at each evaluation, so that the function `brentq` searches over is deterministic – bisection itself assumes a reproducible function, which Eq. 4’s own first attempt implicitly violated by drawing new seeds at every evaluation. $m_1 \in [-0.1, 0.1]$ is p ’s own original grid range from the companion note’s own architecture search (§3.2) – used here without narrowing, since a direct check found no admissibility or numerical irregularity anywhere in this range: κ_0 , kurtosis itself, and the simulated return series’ own maximum absolute value all vary smoothly and continuously across the full bound and beyond. If the real target falls outside what $m_1 \in [-0.1, 0.1]$ can reach at either end, m_1 is set to the corresponding bound.

3.4 Confirmation and confidence band

Every reported result is confirmed by averaging 8 independent longer ($T = 20,000$ -day) simulated paths at m_1^* – a single long path proved too noisy on its own to confirm a kurtosis calibration reliably, so confirmation here also averages, unlike the single-path confirmation used elsewhere in this note family. A 90% confidence band is built from $N = 60$ independent replicate paths ($T = 4,000$ days each):

$$\text{CI}_{90\%} = \left[\widehat{Q}_5(\{\widehat{g}_2^{(s_n)}\}_{n=1}^{60}), \quad \widehat{Q}_{95}(\{\widehat{g}_2^{(s_n)}\}_{n=1}^{60}) \right], \quad (7)$$

where $\widehat{Q}_p(\cdot)$ denotes the p -th empirical percentile.

4 Results

4.1 Excess kurtosis: real vs. calibrated ESN

Figure 1 and Table 2 show the outcome of §3.3’s own per-asset bisection: each asset’s own calibrated m_1 , and how its confirmed excess kurtosis (Eq. 1, averaged over 8 long simulated paths as described in §3.4) compares to its real target, with sampling uncertainty shown as a 90% confidence band.

Every one of the 9 assets’ own real excess kurtosis falls inside the calibrated ESN’s own 90% confidence band, with no exceptions (Figure 1). **2 assets, IEX and NASDAQ100, calibrate to m_1 ’s own lower bound, -0.1** – both have the smallest real targets in the bench (6.204 and 6.040), and their confirmed value (7.170 for both) sits slightly above target rather than

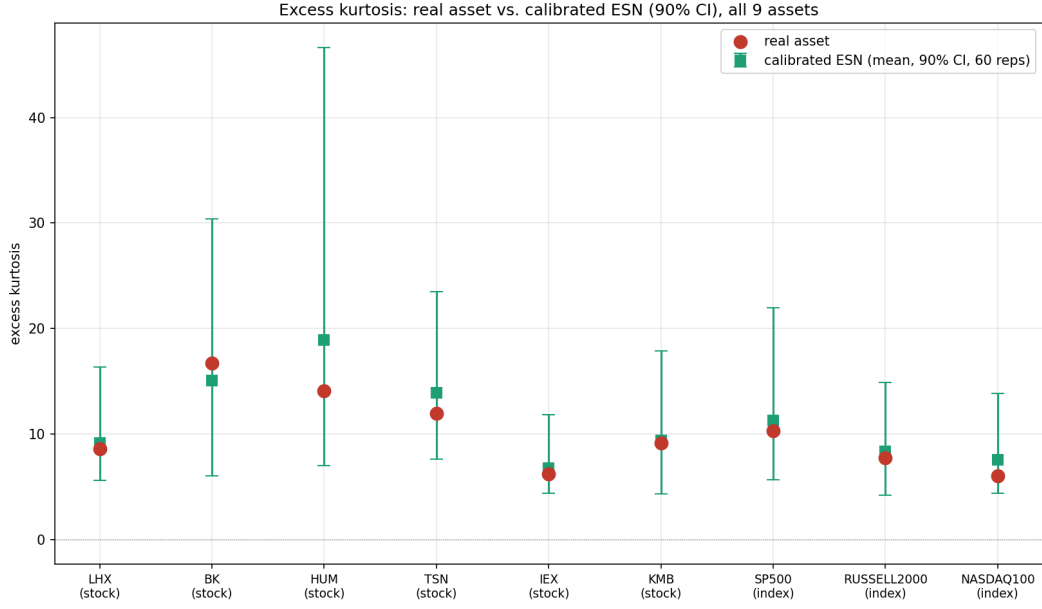


Figure 1: Excess kurtosis: real asset (red) vs. the calibrated ESN's own mean and 90% confidence band (green, 60 replicate $T = 4,000$ -day paths), all 9 assets.

exactly matching it, though still comfortably inside their own wide confidence band. The remaining 7 assets calibrate to an interior m_1 value and match their own target closely.

Table 2: Calibration: fitted m_1 (with $H = 0.3$, `rough_scale` = 0.7, $\lambda_{lo} = 10^{-5}$, $\lambda_{hi} = 1.0$ shared by every asset), all 9 assets.

Asset	type	m_1	confirmed excess kurtosis
LHX	stock	-0.0318	10.431
BK	stock	+0.0330	16.130
HUM	stock	+0.0506	18.384
TSN	stock	+0.0272	15.470
IEX	stock	-0.1000	7.170
KMB	stock	-0.0561	9.038
SP500	index	-0.0155	11.554
RUSSELL2000	index	-0.0724	8.260
NASDAQ100	index	-0.1000	7.170

4.2 Beyond kurtosis: quantile-quantile comparison of the full return distribution

Excess kurtosis is a single, symmetric summary of tail heaviness – it cannot distinguish a genuine match in the full return distribution’s shape from an average of 2 offsetting tail mismatches (a fatter-than-real left tail and a thinner-than-real right one, say). To check this directly, real and calibrated ESN return quantiles are compared at 39 probability levels, denser in the tails ($p \in \{0.001, \dots, 0.999\}$) than the centre, with a 90% confidence band built the same way as §3.3’s own kurtosis band, but now for each individual quantile, across 40 independent replicate paths ($T = 4,000$ days each) at the same calibrated m_1 .

For most assets, the calibrated ESN’s own mean quantile curve tracks $y = x$ closely through the centre of the distribution and for moderate quantiles, consistent with §3.3’s finding that overall kurtosis is well matched. **At the most extreme quantiles, several assets show a visible, asymmetric deviation from $y = x$** – most clearly SP500 and HUM, where the calibrated ESN’s own left-tail quantiles sit noticeably below the $y = x$ line (a fatter left tail than real data shows at that same probability level) while the right tail tracks more closely. **This is a genuinely different check than kurtosis alone can make**: a single scalar summary cannot see whether a good overall match is actually 2 tails matching individually, or one tail overshooting while the other undershoots by a similar amount – the same caution the Taylor-effect and Zumbach-effect notes’ own component-level checks raised for their own single-scalar and difference-based objectives.

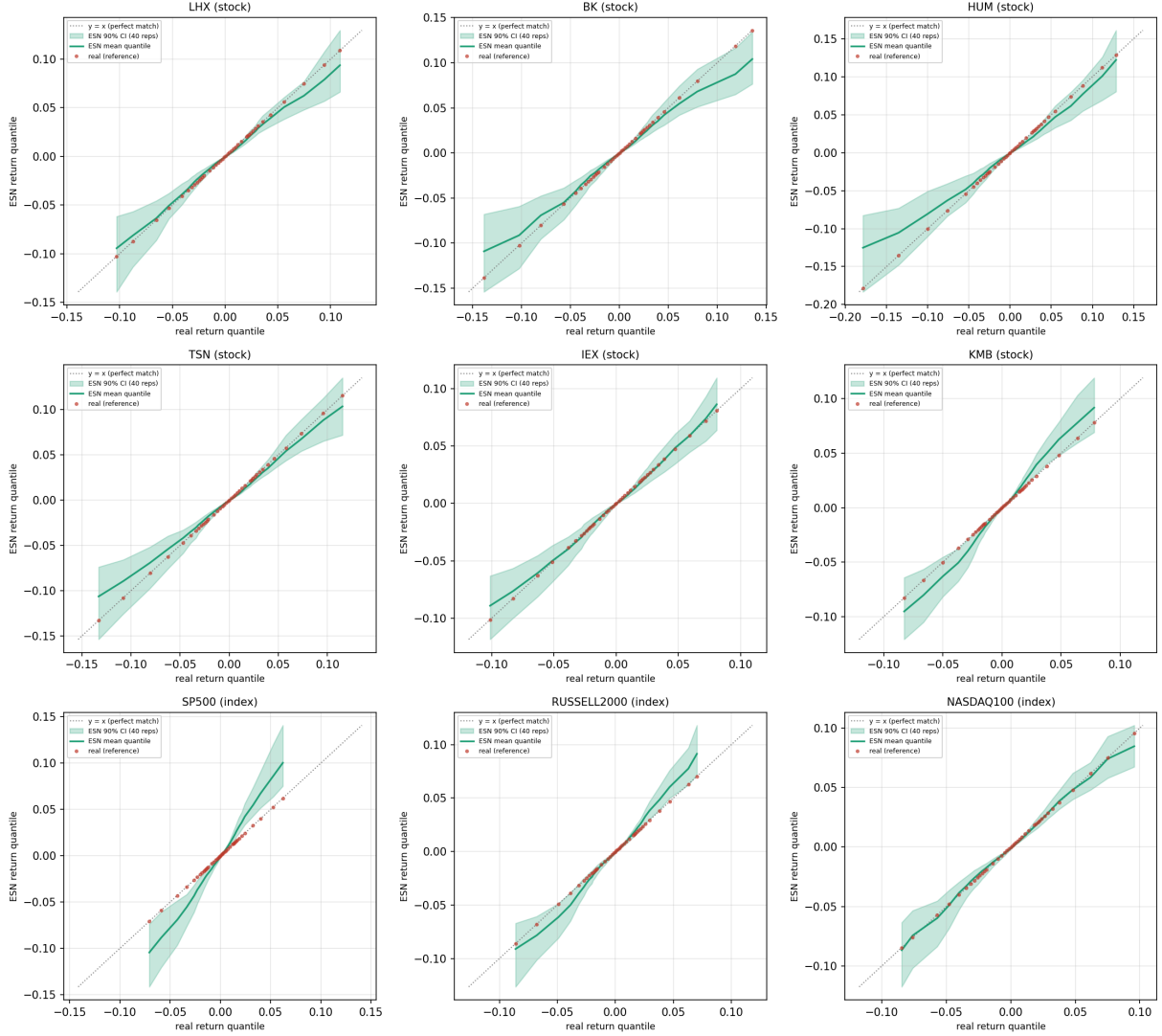


Figure 2: Quantile-quantile plot: real return quantile (horizontal axis) vs. the calibrated ESN's own quantile at the same probability level (vertical axis) – mean (green) and 90% confidence band (shaded, 40 replicate $T = 4,000$ -day paths) – against the $y = x$ reference line (dotted) a perfect distributional match would follow, all 9 assets.

4.3 The full return distribution, directly

Figure 3 shows the same comparison a third way: the full empirical density of standardised returns (each series shifted to zero mean and scaled to unit standard deviation), real asset against a large pooled sample from the calibrated ESN (40 replicate paths, $T = 4,000$ days each, 160,000 points in total per asset – far more than any single real series, so the ESN’s own histogram is a much smoother, more stable estimate than the real one, which is comparatively noisy this far into its own tails). Plotted on a log density scale so that agreement or disagreement in the tails, not just the bulk, is visible directly.

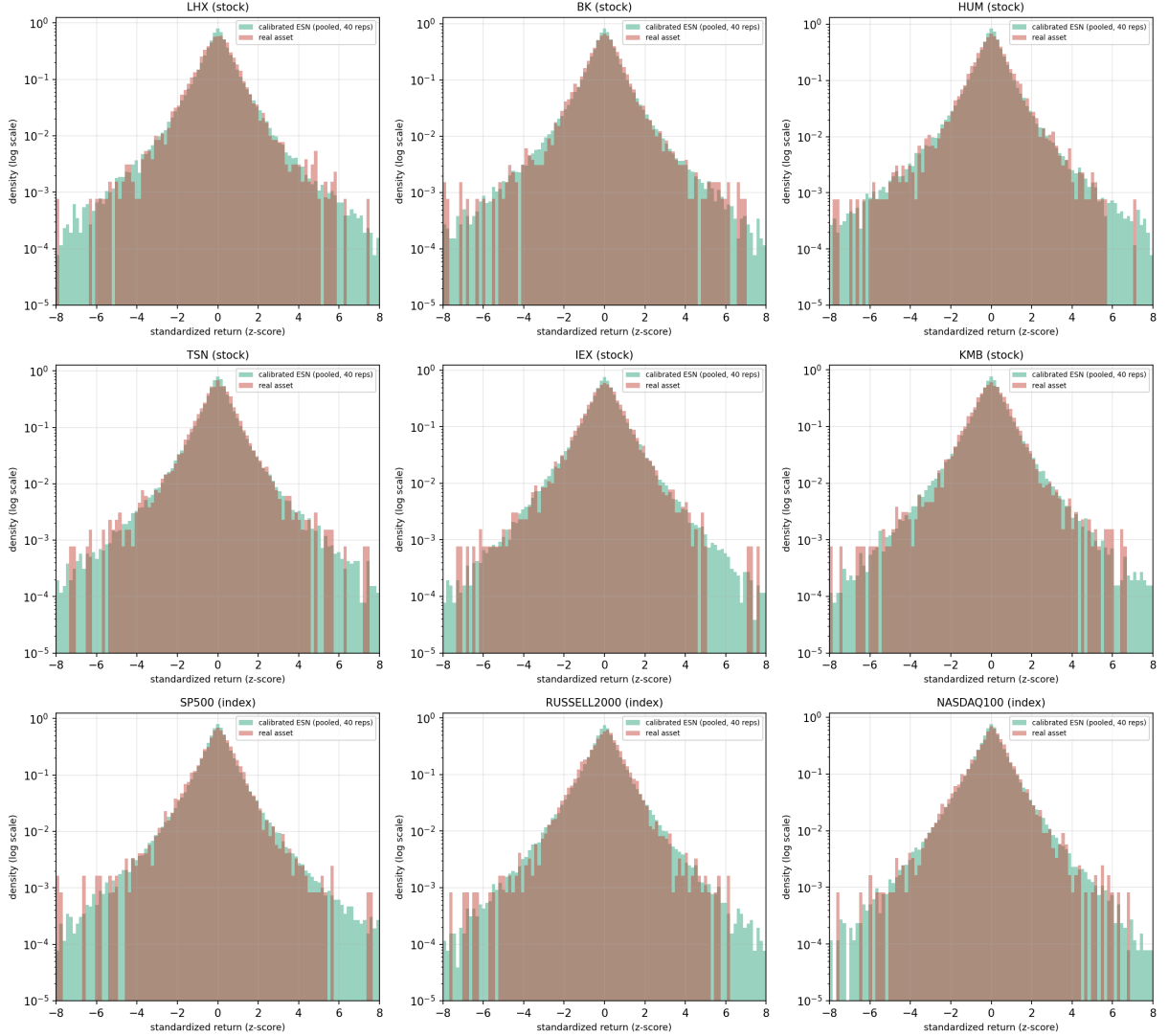


Figure 3: Standardised return distribution (log density scale): real asset (red) vs. the calibrated ESN’s own pooled sample (green, 40 replicate $T = 4,000$ -day paths), all 9 assets.

The 2 distributions overlap closely across their full range for every asset, including several orders of magnitude down into the tails – a direct visual confirmation of §4.2’s own quantile-level finding that the centre and moderate quantiles match well. The asymmetric tail mismatch §4.2 identifies for SP500 and HUM is visible here too on close inspection (the green histogram’s own left tail

extends slightly beyond the red one at the same density level for those 2 assets specifically), but is subtle at this view’s own resolution compared to the QQ plot’s more sensitive, quantile-by-quantile comparison.

5 Interpretation

The heavy-tailedness gap, like the front-loading, persistence ceiling, and remaining offsets found in this note family’s other 3 notes, turned out to be reachable within the existing 4-parameter family plus one already-defined but never-freed parameter – not a fundamentally missing mechanism. Unlike those 3 notes, though, the relevant lever here is m_1 , not H or λ_{hi} : the same ”good corner” that fixed leverage, the Taylor effect, and the Zumbach effect already supplies more kurtosis than most real assets need, and m_1 provides clean, bidirectional control around that baseline. **The optimisation itself needed a different approach here, too:** kurtosis’s own high estimation variance defeated a direct 5-dimensional gradient search that had worked for every other stylised fact in this note family, and fixing 4 parameters to solve a well-behaved 1-dimensional problem for the 5th was the more reliable path, not a lesser one.

6 Limitations and Recommended Next Steps

1. H , `rough_scale`, λ_{lo} , λ_{hi} were fixed at a single shared corner for every asset, rather than calibrated individually. Whether allowing these 4 to vary per asset alongside m_1 would improve the fit further, once the underlying 5-dimensional optimisation difficulty (§3.1) is addressed some other way (e.g. many more paths per evaluation, or a derivative-free method better suited to a noisy objective), was not tested.
2. IEX and NASDAQ100 calibrate to m_1 ’s own lower bound, -0.1 , without exactly reaching their own (lowest) real targets. Whether a wider lower bound would let m_1 push further and close this fully, following the same ”the bound, not the model, was limiting” pattern found for H and λ_{hi} in this note family’s other documents, was not tested here.
3. m_2 , the other quadratic coupling term in η_t , held at its own midpoint of 0 throughout, was not tested as an alternative or additional lever. A brief sweep (in this note’s own supporting code) found m_2 ’s own effect on kurtosis much weaker than m_1 ’s, but a joint m_1, m_2 calibration was not attempted.
4. The confidence bands are wide (Figure 1 shows some spanning close to an order of magnitude), reflecting kurtosis’s own genuinely high estimation variance rather than a calibration weakness – but this also means the calibration’s own precision cannot be judged as tightly as for the other stylised facts in this note family.
5. The asymmetric tail mismatch §4.2’s QQ plot reveals for SP500 and HUM specifically was not resolved. m_1 moves both tails together (it enters η_t through $\|r_t\|^2$, an even function with no left/right asymmetry of its own); a genuinely different lever – e.g. allowing `rough_orientation`’s own sign to vary, or freeing m_2 alongside m_1 – would likely be needed to separate the 2 tails, and was not tested here.
6. The 6 stocks and 3 indices were not chosen for any particular economic or sectoral reason – the stocks are an independently randomised selection (seed 2026), and the indices are the only 3 with available OHLC data.